COER UNIVERSITY

DBMS LAB ASSIGNMENT | LAB SHEET 4: SQL OPERATORS

1. DATABASE ENVIRONMENT & SAMPLE DATA SEEDING

To execute the queries and display correct output tables inside MySQL Workbench, the following database schema and mock records are generated and used as the contextual foundation:

```
-- Step 1: Create Database and Tables
CREATE DATABASE IF NOT EXISTS Lab4_Operators;
USE Lab4_Operators;

CREATE TABLE IF NOT EXISTS Product (
    ProductID INT PRIMARY KEY AUTO_INCREMENT,
    ProductName VARCHAR(50),
    Price DECIMAL(10,2),
    Quantity INT
);

CREATE TABLE IF NOT EXISTS Student (
    StudentID INT PRIMARY KEY AUTO_INCREMENT,
    Name VARCHAR(50),
    Course VARCHAR(50),
    City VARCHAR(50),
    Fees DECIMAL(10,2)
);

CREATE TABLE IF NOT EXISTS Employee (
    EmpID INT PRIMARY KEY AUTO_INCREMENT,
    Name VARCHAR(50),
    Department VARCHAR(50),
    Salary DECIMAL(10,2),
    City VARCHAR(50)
);

-- Step 2: Populate Data
INSERT INTO Product (ProductName, Price, Quantity) VALUES
('Laptop', 65000.00, 2), ('Mouse', 1500.00, 5), ('Keyboard', 2500.00, 3), ('Monitor', 12000.00, 1);

INSERT INTO Student (Name, Course, City, Fees) VALUES
('Amit Sharma', 'B.Tech CSE', 'Roorkee', 75000.00), ('Anjali Rai', 'B.Tech CSE', 'Delhi', 80000.00),
('Akash Verma', 'B.Tech ECE', 'Roorkee', 60000.00), ('Deepak Kumar', 'B.Com', NULL, 45000.00);

INSERT INTO Employee (Name, Department, Salary, City) VALUES
('Rahul Dev', 'IT', 58000.00, 'Delhi'), ('Sonia Malhotra', 'HR', 52000.00, 'Roorkee'),
('Vikas Singh', 'IT', 70000.00, 'Delhi'), ('Anand Mehta', 'Finance', 65000.00, 'Delhi');
```

2. ARITHMETIC OPERATORS

Question 1: Perform addition of two numbers using +.

Explanation: *Adds two numeric values.*

```
SELECT 25 + 15 AS AdditionResult;
```

Result Grid Output (MySQL Workbench Format):

AdditionResult
40

Question 2: Perform subtraction using -.

Explanation: *Subtracts second operand from first.*

```
SELECT 50 - 18 AS SubtractionResult;
```

Result Grid Output (MySQL Workbench Format):

SubtractionResult
32

Question 3: Perform multiplication using *.

Explanation: *Multiplies two numbers.*

```
SELECT 12 * 8 AS MultiplicationResult;
```

Result Grid Output (MySQL Workbench Format):

MultiplicationResult
96

Question 4: Perform division using /.

Explanation: *Divides numerator by denominator.*

```
SELECT 100 / 4 AS DivisionResult;
```

Result Grid Output (MySQL Workbench Format):

DivisionResult
25.0000

Question 5: Calculate the total amount as Price * Quantity.

Explanation: *Multiplies Price and Quantity columns for each row.*

```
SELECT ProductName, Price, Quantity, (Price * Quantity) AS TotalAmount FROM Product;
```

Result Grid Output (MySQL Workbench Format):

ProductName	Price	Quantity	TotalAmount
Laptop	65000.00	2	130000.00
Mouse	1500.00	5	7500.00
Keyboard	2500.00	3	7500.00
Monitor	12000.00	1	12000.00

2. LOGICAL OPERATORS

Question 6: Demonstrate AND with two conditions.

Explanation: *Filters rows where BOTH conditions are TRUE.*

```
SELECT * FROM Student WHERE City = 'Delhi' AND Fees > 65000;
```

Result Grid Output (MySQL Workbench Format):

StudentID	Name	Course	City	Fees
2	Anjali Rai	B.Tech CSE	Delhi	80000.00

Question 7: Demonstrate OR with two conditions.

Explanation: *Filters rows where EITHER condition is TRUE.*

```
SELECT * FROM Student WHERE City = 'Delhi' OR City = 'Roorkee';
```

Result Grid Output (MySQL Workbench Format):

StudentID	Name	Course	City	Fees
1	Amit Sharma	B.Tech CSE	Roorkee	75000.00
2	Anjali Rai	B.Tech CSE	Delhi	80000.00
3	Akash Verma	B.Tech ECE	Roorkee	60000.00

Question 8: Demonstrate NOT with a condition.

Explanation: *NOT negates the truth value of the expression.*

```
SELECT * FROM Student WHERE NOT (City = 'Delhi');
```

Result Grid Output (MySQL Workbench Format):

StudentID	Name	Course	City	Fees
1	Amit Sharma	B.Tech CSE	Roorkee	75000.00
3	Akash Verma	B.Tech ECE	Roorkee	60000.00

Question 9: Find students belonging to a specific course AND city.

Explanation: *Applies AND operator across Course and City columns.*

```
SELECT * FROM Student WHERE Course = 'B.Tech CSE' AND City = 'Roorkee';
```

Result Grid Output (MySQL Workbench Format):

StudentID	Name	Course	City	Fees
1	Amit Sharma	B.Tech CSE	Roorkee	75000.00

Question 10: Find employees belonging to either of two departments.

Explanation: *Selects employees in IT or HR departments.*

```
SELECT * FROM Employee WHERE Department = 'IT' OR Department = 'HR';
```

Result Grid Output (MySQL Workbench Format):

EmpID	Name	Department	Salary	City
1	Rahul Dev	IT	58000.00	Delhi
2	Sonia Malhotra	HR	52000.00	Roorkee
3	Vikas Singh	IT	70000.00	Delhi

2. COMPARISON OPERATORS

Question 11: Demonstrate the = operator.

Explanation: *Checks exact string equality.*

```
SELECT * FROM Student WHERE City = 'Delhi';
```

Result Grid Output (MySQL Workbench Format):

StudentID	Name	Course	City	Fees
2	Anjali Rai	B.Tech CSE	Delhi	80000.00

Question 12: Demonstrate the > operator.

Explanation: *Filters employees earning strictly more than 55,000.*

```
SELECT * FROM Employee WHERE Salary > 55000.00;
```

Result Grid Output (MySQL Workbench Format):

EmpID	Name	Department	Salary	City
1	Rahul Dev	IT	58000.00	Delhi
3	Vikas Singh	IT	70000.00	Delhi
4	Anand Mehta	Finance	65000.00	Delhi

Question 13: Demonstrate the < operator.

Explanation: *Filters products priced below 2,000.*

```
SELECT * FROM Product WHERE Price < 2000.00;
```

Result Grid Output (MySQL Workbench Format):

ProductID	ProductName	Price	Quantity
2	Mouse	1500.00	5

Question 14: Demonstrate >= and <=.

Explanation: *Tests inclusive lower and upper salary bounds.*

```
SELECT * FROM Employee WHERE Salary >= 50000.00 AND Salary <= 65000.00;
```

Result Grid Output (MySQL Workbench Format):

EmpID	Name	Department	Salary	City
1	Rahul Dev	IT	58000.00	Delhi
2	Sonia Malhotra	HR	52000.00	Roorkee
4	Anand Mehta	Finance	65000.00	Delhi

Question 15: Demonstrate the <>/!= operator.

Explanation: *Filters rows where City is not equal to 'Delhi'.*

```
SELECT * FROM Student WHERE City != 'Delhi';
```

Result Grid Output (MySQL Workbench Format):

StudentID	Name	Course	City	Fees
1	Amit Sharma	B.Tech CSE	Roorkee	75000.00
3	Akash Verma	B.Tech ECE	Roorkee	60000.00

2. SPECIAL OPERATORS

Question 16: Demonstrate BETWEEN for salary range.

Explanation: *Selects values within an inclusive range.*

```
SELECT * FROM Employee WHERE Salary BETWEEN 50000.00 AND 65000.00;
```

Result Grid Output (MySQL Workbench Format):

EmpID	Name	Department	Salary	City
1	Rahul Dev	IT	58000.00	Delhi
2	Sonia Malhotra	HR	52000.00	Roorkee
4	Anand Mehta	Finance	65000.00	Delhi

Question 17: Demonstrate IN for selecting multiple cities.

Explanation: *IN tests if a column value matches any item in a discrete list.*

```
SELECT * FROM Student WHERE City IN ('Delhi', 'Roorkee', 'Dehradun');
```

Result Grid Output (MySQL Workbench Format):

StudentID	Name	Course	City	Fees
1	Amit Sharma	B.Tech CSE	Roorkee	75000.00
2	Anjali Rai	B.Tech CSE	Delhi	80000.00
3	Akash Verma	B.Tech ECE	Roorkee	60000.00

Question 18: Demonstrate NOT IN.

Explanation: *Excludes students from cities in the list.*

```
SELECT * FROM Student WHERE City NOT IN ('Delhi', 'Meerut');
```

Result Grid Output (MySQL Workbench Format):

StudentID	Name	Course	City	Fees
1	Amit Sharma	B.Tech CSE	Roorkee	75000.00
3	Akash Verma	B.Tech ECE	Roorkee	60000.00

Question 19: Demonstrate LIKE for names beginning with a particular letter.

Explanation: *LIKE 'A%' matches string values beginning with 'A'.*

```
SELECT * FROM Student WHERE Name LIKE 'A%';
```

Result Grid Output (MySQL Workbench Format):

StudentID	Name	Course	City	Fees
1	Amit Sharma	B.Tech CSE	Roorkee	75000.00
2	Anjali Rai	B.Tech CSE	Delhi	80000.00
3	Akash Verma	B.Tech ECE	Roorkee	60000.00

Question 20: Demonstrate IS NULL and IS NOT NULL.

Explanation: *Tests for missing (NULL) or present values.*

```
SELECT * FROM Student WHERE City IS NULL;
SELECT * FROM Student WHERE City IS NOT NULL;
```

Result Grid Output (MySQL Workbench Format):

StudentID	Name	Course	City	Fees
4	Deepak Kumar	B.Com	NULL	45000.00

2. SET OPERATIONS & INTEGRATED QUERIES

Question 21: Demonstrate UNION using two compatible queries.

Explanation: *UNION combines distinct results from two queries into a single result set.*

```
SELECT Name, City FROM Student UNION SELECT Name, City FROM Employee;
```

Result Grid Output (MySQL Workbench Format):

Name	City
Amit Sharma	Roorkee
Anjali Rai	Delhi
Akash Verma	Roorkee
Deepak Kumar	NULL
Rahul Dev	Delhi
Sonia Malhotra	Roorkee
Vikas Singh	Delhi
Anand Mehta	Delhi

Question 23: Demonstrate INTERSECT where supported by the DBMS.

Explanation: *Returns rows that appear in both result sets.*

```
SELECT City FROM Student INTERSECT SELECT City FROM Employee;
```

Result Grid Output (MySQL Workbench Format):

City
Roorkee
Delhi

Question 24: Demonstrate EXCEPT/MINUS where supported.

Explanation: *Returns distinct rows from the first query that do not appear in the second.*

```
SELECT City FROM Student EXCEPT SELECT City FROM Employee;
```

Result Grid Output (MySQL Workbench Format):

City
NULL

Question 25: Perform a problem using Arithmetic + Logical + Comparison + Special operators.

Explanation: *Combines arithmetic (*), comparison (>), logical (AND), and special (IN, IS NOT NULL) operators.*

```
SELECT EmpID, Name, Salary, (Salary * 12) AS AnnualSalary FROM Employee
WHERE (Salary * 12) > 600000
AND Department IN ('IT', 'Finance')
AND City = 'Delhi'
AND Salary IS NOT NULL;
```

Result Grid Output (MySQL Workbench Format):

EmpID	Name	Salary	AnnualSalary
1	Rahul Dev	58000.00	696000.00
3	Vikas Singh	70000.00	840000.00
4	Anand Mehta	65000.00	780000.00

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